

Revised criteria for the diagnosis and staging of Alzheimer's disease

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Alzheimer's disease can be treated by targeting amyloid- β plaques and diagnosed in vivo by biomarkers, prompting the revision of criteria for the diagnosis and staging of this disease.

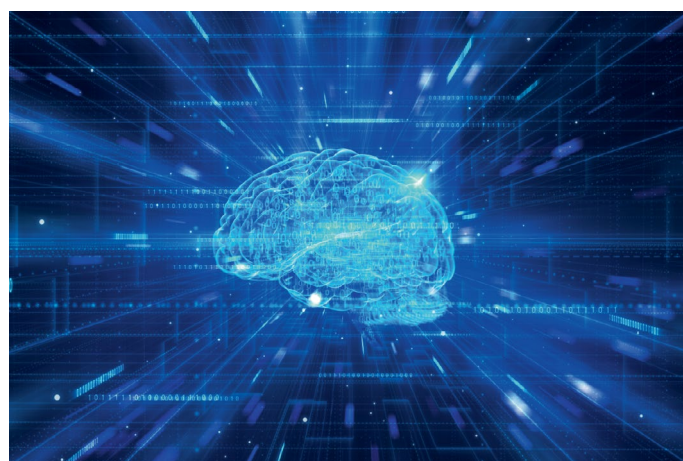
A workgroup convened by the Alzheimer's Association recently published 'Revised Criteria for Diagnosis and Staging of Alzheimer's Disease'¹. Members of this committee (who also co-authored this comment) were selected to provide balanced representation of academia, clinical medicine, industry, the US National Institute on Aging (NIA) and the US Food and Drug Administration (FDA), and included participants from the USA and Europe. The revised criteria¹ are an update to previous criteria published in 2011 and 2018^{2,3}. Formulation of these revised criteria¹ was prompted by two major developments in this field that occurred after the 2018 research framework³ was published: regulatory approval of the first disease-targeted therapeutic agents for Alzheimer's disease^{4,5}; and development of accurate blood-based biomarkers of the disease⁶.

The 2024 revised criteria¹ are founded on several core principles. Alzheimer's disease is defined as a biological process that is first detectable by abnormal biomarkers of its defining neuropathologic features (amyloid- β plaques and tau neurofibrillary tangles) when an individual is asymptomatic. The disease progresses biologically during the pre-clinical period and when a sufficient pathological burden is reached, symptoms appear and then progress. The entire disease course may span up to 30 years.

Categorization of biomarkers

In the revised criteria¹, biomarkers are categorized into those that are core biomarkers of neuropathological changes⁷ (specific to Alzheimer's disease), non-core biomarkers of processes (not specific to Alzheimer's but still important in disease pathogenesis, such as biomarkers of inflammation or immune activation, and neurodegeneration), and biomarkers of common non-Alzheimer's disease co-pathologies (such as cerebrovascular disease and neuronal α -synuclein disease). This commentary focuses on core biomarkers for Alzheimer's disease that fall into the following categories: biomarkers of the amyloid- β proteinopathy pathway (labeled 'A'); biofluid biomarkers of phosphorylated and secreted tau, which become abnormal early in the disease process (labeled 'T₁'); and biomarkers of tau neurofibrillary pathology that become abnormal later in the disease process (labeled 'T₂'). We group biomarkers into core 1 (A and T₁) and core 2 (T₂) (Table 1).

Core 1 biomarkers define the initial stage of Alzheimer's disease that is detectable in vivo. An abnormal core 1 biomarker is sufficient



to establish a diagnosis of Alzheimer's disease, and to inform clinical decision making (Table 2). Core 2 biomarkers are not typically used as standalone tests for the diagnosis of Alzheimer's disease, but when combined with core 1 they may be used to stage biological disease severity and provide prognostic information (Table 2).

Diagnosis

An abnormal core 1 biomarker test is diagnostic of Alzheimer's disease. The following are currently considered fit for this purpose: amyloid positron emission tomography (PET); the cerebrospinal fluid (CSF) ratio of A β ₄₂ to A β ₄₀ (A β _{42/40}); the CSF ratio of tau phosphorylated at threonine 181 (p-tau₁₈₁) to A β ₄₂ (p-tau₁₈₁/A β ₄₂); the CSF ratio of total tau to A β ₄₂ (t-tau/A β ₄₂) ratio; and accurate plasma assays (defined below). However, the revised criteria outline important caveats around diagnostics, especially using plasma biomarkers. Chief among these is a minimum diagnostic accuracy requirement (90% or greater) with respect to an accepted reference standard in the intended context of use (which includes a description of the intended use population). Amyloid PET ligands and CSF assays listed in Table 1 have received regulatory approval in the USA and the EU; the former were validated against autopsy⁸ and the latter were validated against amyloid PET⁹. Although no plasma assays have yet received regulatory approval, this is expected to change soon as some plasma assays (such as p-tau₂₁₇) have demonstrated diagnostic accuracy equivalent to approved CSF assays¹⁰. The diagnostic accuracies of other plasma assays, however, do not meet the 90% standard¹⁰ and would not be considered fit for the purpose of diagnosis.

A second important caveat centers on the role of the clinician. The biologically based diagnosis of Alzheimer's disease is meant to

Table 1 | Categorization of core Alzheimer’s disease biomarkers

Category	Modalities
Core 1	Amyloid PET and biofluid measures of amyloid-β proteinopathy (A)
	Biofluid measures of phosphorylated and secreted Alzheimer’s disease tau (T ₁)
Core 2	Tau PET and biofluid measures of Alzheimer’s disease tau proteinopathy (T ₂)

assist, rather than supplant, the clinical evaluation of individuals with cognitive impairment. Biomarker testing should only be performed under the supervision of a clinician.

Staging

The revised criteria¹ describe separate schemes for staging biological Alzheimer’s disease severity and for staging the severity of clinical symptoms. Biological staging is based on ordering of biomarker events in the natural history of the disease from observational research. A four-point staging scale (A–D) is outlined that can be accomplished by amyloid and tau PET, by core 1 biofluids and tau PET, or a forward-looking conceptual staging approach based only on biofluids^{11,12}. Staging the severity of clinical symptoms is based on a six-point numeric scale ranging from asymptomatic (stage 1) to severe dementia (stage 6). Integration of biological and clinical staging is described using an alphanumeric labeling scheme (for example, stage 1A, 4D, and so on).

Although clinical stages generally worsen with increasing biological stages, the revised criteria recognize that mismatches between clinical and biological stage commonly occur. This is because Alzheimer’s disease is only one of several common pathologies that underlie cognitive decline and dementia in older individuals; the other three most common disorders are cerebrovascular disease, neuronal α-synuclein disease, and limbic-predominant age-related TDP-43 encephalopathy (LATE)¹³. Individuals with one or more of these pathologies are likely to have worse clinical stage than expected for their biological Alzheimer’s disease stage. Conversely, individuals with exceptional resilience or cognitive reserve may have better clinical stage than expected for their biological Alzheimer’s disease stage.

Use of the name Alzheimer’s disease

Two successive drafts of these criteria were presented at scientific meetings in 2023 and were also posted for public comment. It is our view that some of the comments or critiques offered contradict fundamental positions of the committee and so below we describe these critiques and the rational underlying the committee’s decision to formulate the criteria as done.

The term Alzheimer’s disease is understood by many in the public and in general medicine to be synonymous with dementia. Many people think of Alzheimer’s disease as a progressive impairment that leads to loss of functional independence in older people, irrespective of etiology. A common critique of the revised criteria¹ is that to avoid confusion on the part of the public, the term Alzheimer’s disease should be used to describe all-cause dementia and the distinction between etiology (pathobiology) and symptoms (the result of pathobiology) should be ignored.

The position of the committee is that not only is this scientifically inaccurate, but it is also harmful. The etiology or etiologies underlying clinical symptoms must be understood and accurately diagnosed

Table 2 | Intended uses of core Alzheimer’s disease biomarkers

Category	Intended use	Details
Core 1	Diagnosis	Early detection of Alzheimer’s disease in asymptomatic individuals
		Confirmation that Alzheimer’s disease is an underlying pathology in symptomatic individuals
Core 2	Staging and prognosis	Provide information on the likelihood that symptoms are associated with Alzheimer’s disease
		Inform on the risk of and rate of progression

to enable effective treatments. Although Alzheimer’s disease is the most common pathological condition underlying progressive cognitive impairment in older people, non-Alzheimer’s disease pathologies are also common¹³. The typical syndrome association with Alzheimer’s disease (a progressive memory impairment that progresses to other cognitive domains) can also be caused by common non-Alzheimer’s disease pathologies, most often in combination. For example, in early amyloid-β immunotherapy trials in which eligibility was based on a traditional clinical diagnosis of Alzheimer’s disease without biological confirmation, 30% of individuals had normal amyloid PET scans¹⁴, meaning that nearly one-third of participants in these trials did not have the disease they were being treated for. The position of the committee is that therapy directed toward a biological target requires confirmation of that biology both in clinical trials and in clinical practice. Thus, Alzheimer’s disease must be defined by its biology for medical progress to be made; the public is readily able to grasp this concept with the proper educational effort.

Requiring symptoms for diagnosis

A second critique argues that a diagnosis of Alzheimer’s disease should require the presence of both abnormal biomarkers and the presence of clinical symptoms. The argument is that preclinical Alzheimer’s disease is not a valid concept and asymptomatic individuals with abnormal biomarkers should be labeled ‘at risk’ for the disease. The reasoning is that many asymptomatic individuals with abnormal biomarkers (preclinical Alzheimer’s disease) will not develop symptoms in their lifetime. The argument continues, labeling such individuals with a disease may cause psychological, financial or social harm without potential for benefit, since at present, disease targeted treatments are not approved for preclinical Alzheimer’s disease.

The committee takes a different view¹. First, diseases are routinely diagnosed in other areas of medicine while patients are asymptomatic. It is axiomatic throughout medicine that treatments are more effective when administered as early in the disease process as possible². Therefore, conducting clinical trials in people with preclinical Alzheimer’s disease is considered by the committee to be a forward-looking approach to identify the most effective means of slowing or preventing the onset of symptoms.

Second, although it is true that many asymptomatic individuals with abnormal Alzheimer’s biomarkers will not experience symptoms in their lifetime, this is because of increasing all-cause mortality rates with advancing age, not because Alzheimer’s disease pathology is benign in the preclinical period. Alzheimer’s disease is characterized by a pre-clinical period of 15–20 years or more, which most often begins from age 60¹. Over half of individuals newly diagnosed in their mid-70s may not experience symptoms in their lifetime, but nearly half will. The position

of the committee is that asymptomatic individuals who may experience symptoms deserve treatment once these are approved for the preclinical population. Mortality from unrelated diseases should not be interpreted to indicate that Alzheimer's disease in asymptomatic persons is a benign condition. Individuals with Down syndrome have genetically determined Alzheimer's disease (trisomy 21) and 95% of these individuals develop dementia in their lifetime¹⁵, because the average age of onset of clinical symptoms is mid-50s, at which age all-cause mortality rates are far lower than in older age. Remaining life expectancy is an important consideration in clinical management, but mortality from unrelated causes should not be a criterion used to define what is and what is not a disease¹.

Third, the revised criteria have been criticized for advocating screening of the general population with Alzheimer's disease biomarkers, which could place asymptomatic persons who test positive in a position of potential harm without benefit. However, the criteria are very clear that because no treatment has yet been approved for asymptomatic people, biomarker testing should not be performed in this population outside the context of observational or therapeutic research studies¹. This may change in the future, however, pending results of ongoing trials.

Validation of plasma biomarkers

Criticism was raised that plasma biomarkers are too new to be used clinically and have not been adequately tested in all representative populations. While the committee recognizes that plasma biomarkers are a recent development, we also outline rigorous criteria that plasma biomarkers need to meet to be considered fit for the purpose of diagnosis¹. The committee strongly endorses the need for testing and prospective evaluation of plasma biomarkers in more representative populations.

Neuropathological underpinnings of diagnosis by biomarkers

Some critics claim that defining Alzheimer's disease by an abnormal core 1 biomarker is a departure from the accepted neuropathological definition that requires the presence of both amyloid- β plaques and tau tangles⁷. Regulatory approval of amyloid PET ligands was based on their ability to detect moderate to frequent neuritic amyloid- β plaques at autopsy and not both plaques and tangles⁸, whereas approval of CSF assays was based on their ability to differentiate normal from abnormal amyloid PET scans⁹. Thus, the argument of these critics is that defining Alzheimer's disease by the presence of abnormal core 1 biomarkers does not fully capture the neuropathological standard that requires tau tangles in addition to β -amyloid plaques. However, amyloid PET scans (and thus biofluid surrogates) are not able to reliably detect mild levels of Alzheimer's disease pathology; it can detect moderate to frequent neuritic plaques but not sparse plaques⁸. Given that nearly all (over 90%) of individuals with moderate to frequent neuritic plaques at autopsy (and therefore abnormal core 1 biomarkers) will also have sufficient neurofibrillary tangle pathology to meet criteria for a pathological diagnosis of intermediate to high Alzheimer's disease neuropathologic change⁷, diagnosing Alzheimer's disease based on abnormal core 1 biomarkers will nearly always be consistent with the accepted neuropathologic reference standard for Alzheimer's disease.

Future

Many in the Alzheimer's disease field are optimistic about the prospects for improving care of patients. Recent advances in the ability to diagnose and treat Alzheimer's disease herald a virtuous cycle in

which improvements in diagnostic methods enable more sophisticated treatment approaches, which in turn steer advances in diagnostic methods. An unchanging principle, however, is that effective treatment will always rely on the ability to diagnose and stage the biology driving the disease process. The committee recognizes that the feasibility of implementing criteria for biologically based diagnosis and staging of Alzheimer's disease in clinical practice varies across regions, even within high-income countries. We anticipate that the increasing availability and accuracy of blood-based biomarkers will make these criteria more widely deployable.

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Competing interests

C.R.J. is employed by Mayo Clinic; receives grant funding from the National Institutes of Health (NIH), the Alexander family professorship and the GHR Foundation; served on a data and safety monitoring board for Roche pro bono within the past 36 months with no payments to the individual or institution involved; and received funding from the Alzheimer's Association for travel and holds index funds. J.S.A. is employed by Takeda Pharmaceuticals and is a minor shareholder for the company; has a leadership or fiduciary role in the Clinical Trials on Alzheimer's Disease (CTAD) Task Force—Clinical Meaningfulness and Optimizing Therapies, Alzheimer's Disease PACE Executive Steering Committee and UsAgainstAlzheimer's—Clinical Meaningfulness Forum; is a former employee of Eli Lilly and Company and is a minor shareholder for the company. T.G.B. is employed by Banner Health; received grant funding from the NIH, Veterans Administration, State of Arizona, Life Molecular Imaging, Michael J. Fox Foundation for Parkinson's Research, Avid Radiopharmaceuticals/Eli Lilly and Gates Foundation; received consulting fees from Aprinoia Therapeutics and Acadia Pharmaceuticals, and consulted unpaid for Biogen; received payment or honoraria from the World PD Coalition, Mayo Clinic Florida, Stanford University and the IOS Press-*Journal of Parkinson's Disease*; received support for attending meetings from the Alzheimer's Association, AD/PD/Kenes Group, Mayo Clinic Florida and the Universitätsklinikum Hamburg-Eppendorf; and has a leadership or fiduciary role and stock options with Vivid Genomics. T.B. is employed by the US FDA and has no financial conflicts to disclose. B.D. is a consultant and received consulting fees for his role as an advisor for Arch Venture Partners, Cerveau Technologies, Epilepsy Foundation, F-PRIME Capital, Loulou Foundation and Michael J. Fox Foundation; held a past position of leadership or fiduciary role in the Virginia Neurological Society (past president) and Prothena (director); and holds stock options with Prothena. A.G. is employed by Novartis Pharma AG and has stock options in the company. O.H. is employed by Lund University; received grants or contracts from ADx, AVID Radiopharmaceuticals, Biogen, Eli Lilly, Eisai, Fujirebio, GE Healthcare, Pfizer and Roche; and received consulting fees from AC Immune, Amylyx, Alzpath, BioArctic, Biogen, Cerveau, Eisai, Eli Lilly, Fujirebio, Genentech, Merck, Novartis, Novo Nordisk, Roche, Sanofi and Siemens. C.H. is employed by Denali Therapeutics and has stock options with the company; is a board director for Beam Therapeutics and NGM Therapeutics and has stock options with both companies. W.J. is employed by the University of California, Berkeley; received grants or contracts paid to his institution from the US NIA, Roche/Genentech, Alzheimer's Association and BrightFocus Foundation; received consulting fees from Biogen, Clario, Eisai, Lilly and Prothena; and has stock with Optoceutics and Molecular Medicine. E. McDade is employed by Washington University, St Louis, and received grant funding directly to his institution from NIA, Eli Lilly, Hoffmann-La Roche, Alzheimer's Association and GHR Foundation; received royalties for work on 'Methods of diagnosing AD with phosphorylation changes', which is licensed to C2N with royalties to himself and Washington University; received consulting fees from Alzamend (scientific advisory board

member), Sanofi, AstraZeneca, Roche, Grifols, Merck and Sage; received payments from Neurology Live, Kaplan-Projects in Knowledge, and support for travel from Foundation Alzheimer, Alzheimer's Association and Eisai; has a patent planned, issued or pending on 'Methods of diagnosing Alzheimer's disease with phosphorylation changes'; participated on a data safety monitoring board or advisory board for Alektor and Eli Lilly, both roles for which he was paid; and has a leadership or fiduciary role in Alzamend (paid). J.L.M. is a full-time employee of H. Lundbeck A/S. O.O. is employed by University of Wisconsin School of Medicine & Public Health and received support paid to his institution from the following grants: R01AG062167, R01AG077507, U19AG024904, U19AG078109, U19AG073153, R01AG066203, R01AG070028, R01AG027161 and RF1AG052324; received consulting fees from Mayo Clinic Rochester and IUPUI; holds a leadership or fiduciary role in the International Neuropsychological Society (board member); and previously held an advisory role for Society for Black Neuropsychology. L.P. is employed by the University of Miami and the University of Modena and Reggio Emilia and has no financial conflicts to disclose; in unrelated areas, he holds options/stocks from Relmada Therapeutics (US) and NetraMark (Canada). M.R. is employed by University of Southern California and the Alzheimer's Therapeutics Research Institute (ATRI); received grants or contracts paid to his institution from Eisai (AHEAD study) and Eli Lilly (A4 study); received consulting fees from AC Immune and Ionis; and has participated on a data safety monitoring board or advisory board for Alzheon, Aptah Bio, Biohaven, Embic, Keystone Bio, Prescient Imaging and Positron. P.S. is employed by Life Science Partners and is professor emeritus Amsterdam University Medical Center; received grants or contracts paid to his institution from Novo Nordisk, Toyama, UCB, AC Immune and Alzheon; has an unpaid leadership position as chair of World Dementia Council and holds stock options in EQT AB. E.S. is employed by Acumen Pharmaceuticals; received consulting fees from Biogen, Cogstate, Cortexyme, Partner Therapeutics, Pinteon Therapeutics, Prothena, Vaccinex, Acumen, Pharmaceuticals, Gates Ventures LLC and Hoffman La Roche and payments were made to Siemers Integration LLC; participated on a data safety monitoring board for Hoffman La Roche; and has had a leadership or fiduciary role with the Alzheimer's Association and BrightFocus Foundation, both unpaid; holds stock options and is a shareholder for Acumen Pharmaceuticals, and is a shareholder for Eli Lilly and Company. H.S. is a full-time employee of the Alzheimer's Association, Chicago, has a spouse who is employed by Abbott Laboratories in an unrelated area, and has no financial conflicts to disclose. R.S. is employed by Brigham and Women's Hospital; received grants or contracts from the US NIA, Eli Lilly (public-private partnership trial funding), Eisai (public-private partnership trial funding), Alzheimer's Association and GHR Foundation; and received consulting fees from Abbvie, AC Immune, Acumen, Alektor, Alnylam, Bristol-Myers Squibb, Cytex, Genentech, Ionis, Janssen, NervGen, Neuraly, Neurocentria, Oligomerix, Prothena, Roche, Shionogi and Vaxxinity. C.E.T. is employed by Amsterdam UMC; received grants or contracts for research from the European Commission (Marie Curie International Training Network, grant 86019, MIRIAD), Innovative Medicines Initiatives 3TR (Horizon 2020, grant 831434) EPND (IMI 2 Joint Undertaking, grant 101034344) and JPND (bPRIDE), National MS Society (Progressive MS Alliance), Alzheimer Drug Discovery Foundation, Alzheimer Association, Health Holland, the Dutch Research Council (ZonMW), including TAP-dementia, a ZonMw funded project (10510032120003) in the context of the Dutch National Dementia Strategy, Alzheimer Drug Discovery Foundation, The Selfridges Group Foundation, Alzheimer Netherlands; is recipient of ABOARD, public-private partnership receiving funding from ZonMW (73305095007) and Health-Holland, Topsector Life Sciences & Health (PPP-allowance; LSHM20106); is a contract researcher for ADx Neurosciences, AC-Immune, Aribio, Axon Neurosciences, Beckman-Coulter, BioConnect, Bioorchestra, Brainstorm Therapeutics, Celgene, Cognition Therapeutics, EIP Pharma, Eisai, Eli Lilly, Fujirebio, Grifols, Instant Nano Biosensors, Merck, Novo Nordisk, Olink, PeopleBio, Quanterix, Roche, Siemens, Toyama, Vivoryon and the European Commission; received payment or honoraria to her institution from Eli Lilly, Grifols, Novo Nordisk, Olink and Roche; serves on editorial boards of *Medidact Neurologie*/Springer and *Neurology: Neuroimmunology & Neuroinflammation*, and is editor of *Alzheimer Research and Therapy*. M.C. is a full-time employee of the Alzheimer's Association, has a daughter in the neuroscience program at the University of South Carolina, and has no financial conflicts to disclose.